

Администрирование сетевых подсистем

Лабораторная работа №2 — Настройка и анализ работы
DNS-сервера

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Цели е задачи работы

Цель лабораторной работы

Приобрести практические навыки установки и конфигурирования DNS-сервера
и закрепить понимание принципов работы системы доменных имён.

Установка е настройка DNS-сервера

Установка пакетов и проверка работы DNS

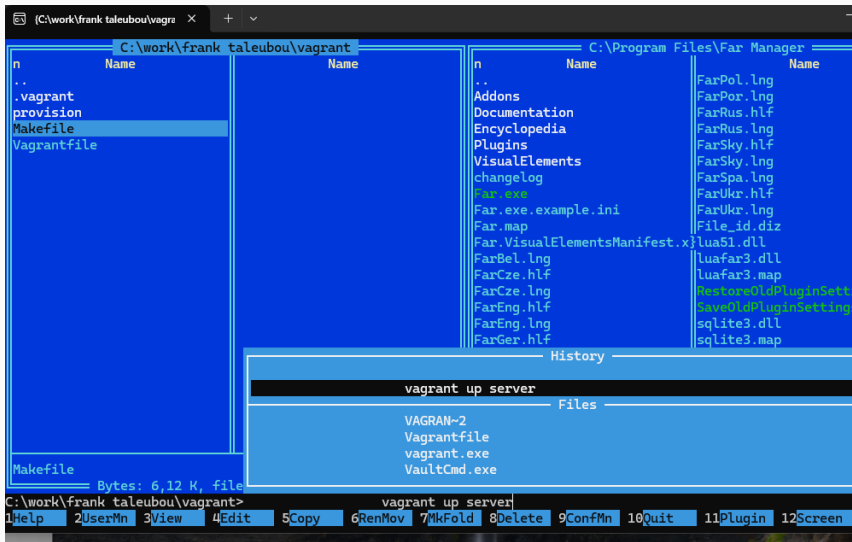


Рис. 1: Результат выполнения команды `dig www.yandex.ru`

Анализ конфигурационных файлов DNS

```
root@server:~  
Verifying      : bind-license-32:9.16.23-34.el9_7.1.noarch  
Verifying      : bind-license-32:9.16.23-18.el9_4.1.noarch  
Verifying      : bind-utils-32:9.16.23-34.el9_7.1.x86_64  
Verifying      : bind-utils-32:9.16.23-18.el9_4.1.x86_64  
Verifying      : openssl-devel-1:3.5.1-7.el9_7.x86_64  
Verifying      : openssl-devel-1:3.0.7-27.el9.x86_64  
  
Upgraded:  
  bind-libs-32:9.16.23-34.el9_7.1.x86_64  
  bind-license-32:9.16.23-34.el9_7.1.noarch  
  bind-utils-32:9.16.23-34.el9_7.1.x86_64  
  openssl-1:3.5.1-7.el9_7.x86_64  
  openssl-devel-1:3.5.1-7.el9_7.x86_64  
  openssl-libs-1:3.5.1-7.el9_7.x86_64  
  
Installed:  
  bind-32:9.16.23-34.el9_7.1.x86_64  
  bind-dnssec-doc-32:9.16.23-34.el9_7.1.noarch  
  bind-dnssec-utils-32:9.16.23-34.el9_7.1.x86_64  
  openssl-fips-provider-1:3.5.1-7.el9_7.x86_64  
  python3-bind-32:9.16.23-34.el9_7.1.noarch  
  python3-ply-3.11-14.el9.0.1.noarch  
  
Complete!  
[root@server ~]#
```

Анализ конфигурационных файлов DNS

```
root@server:~  
[root@server ~]# dig www.yandex.ru  
  
; <<>> DiG 9.16.23-RH <<>> www.yandex.ru  
;; global options: +cmd  
;; Got answer:  
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 37221  
;; flags: qr rd ra; QUERY: 1, ANSWER: 3, AUTHORITY: 0, ADDITIONAL: 1  
  
;; OPT PSEUDOSECTION:  
; EDNS: version: 0, flags:; udp: 4000  
;; QUESTION SECTION:  
;www.yandex.ru.                IN      A  
  
;; ANSWER SECTION:  
www.yandex.ru.                212     IN      A      5.255.255.77  
www.yandex.ru.                212     IN      A      77.88.44.55  
www.yandex.ru.                212     IN      A      77.88.55.88  
  
;; Query time: 24 msec  
;; SERVER: 192.168.150.37#53(192.168.150.37)  
;; WHEN: Wed Mar 04 11:39:44 UTC 2026  
;; MSG SIZE rcvd: 90  
  
[root@server ~]#
```

Анализ конфигурационных файлов DNS

```
root@server:~  
[root@server ~]# dig www.yandex.ru  
  
; <<>> DiG 9.16.23-RH <<>> www.yandex.ru  
;; global options: +cmd  
;; Got answer:  
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 37221  
;; flags: qr rd ra; QUERY: 1, ANSWER: 3, AUTHORITY: 0, ADDITIONAL: 1  
  
;; OPT PSEUDOSECTION:  
; EDNS: version: 0, flags:; udp: 4000  
;; QUESTION SECTION:  
;www.yandex.ru.                IN      A  
  
;; ANSWER SECTION:  
www.yandex.ru.                212     IN      A      5.255.255.77  
www.yandex.ru.                212     IN      A      77.88.44.55  
www.yandex.ru.                212     IN      A      77.88.55.88  
  
;; Query time: 24 msec  
;; SERVER: 192.168.150.37#53(192.168.150.37)  
;; WHEN: Wed Mar 04 11:39:44 UTC 2026  
;; MSG SIZE rcvd: 90  
  
[root@server ~]#
```


Проверка работы DNS через localhost

```
[root@server ~]# nmcli connection edit System\ eth0
==| nmcli interactive connection editor |==
Editing existing '802-3-ethernet' connection: 'System eth0'

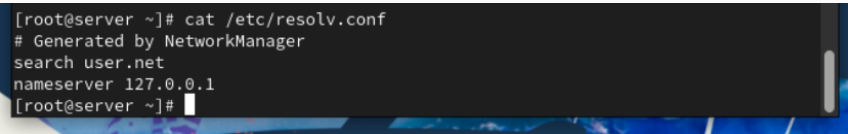
Type 'help' or '?' for available commands.
Type 'print' to show all the connection properties.
Type 'describe [<setting>.<prop>]' for detailed property description.

You may edit the following settings: connection, 802-3-ethernet (ethernet), 802-
1x, dcb, sriov, ethtool, match, ipv4, ipv6, hostname, link, tc, proxy
nmcli> remove ipv4.dns
nmcli> set ipv4.ignore-auto-dns yes
nmcli> set ipv4.dns 127.0.0.1
nmcli> save
Connection 'System eth0' (5fb06bd0-0bb0-7ffb-45f1-d6edd65f3e03) successfully upd
ated.
nmcli> quit
[root@server ~]#
```

Рис. 5: Запрос к DNS через localhost

Конфигурирование первичного DNS-сервера

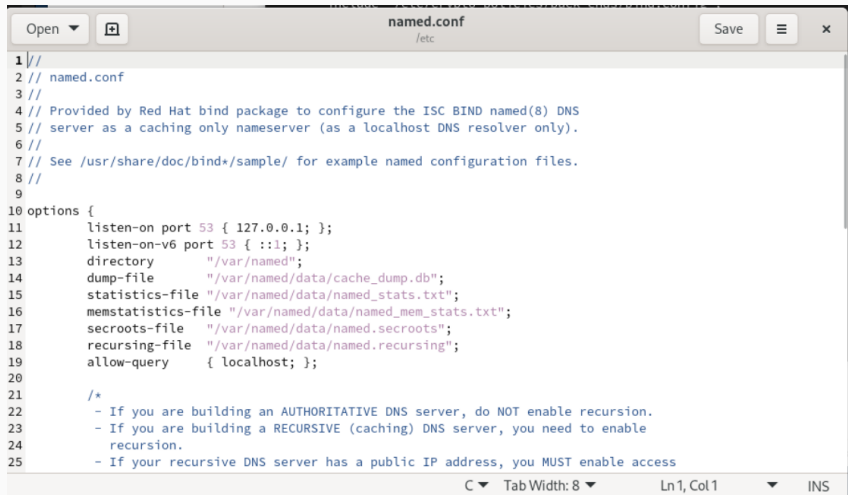
Добавление файла зоны и настройка конфигурации

A terminal window with a dark background and light gray text. The prompt is [root@server ~]#. The command cat /etc/resolv.conf has been executed, and the output is displayed line by line: # Generated by NetworkManager, search user.net, and nameserver 127.0.0.1. The prompt [root@server ~]# is shown again at the end of the output.

```
[root@server ~]# cat /etc/resolv.conf
# Generated by NetworkManager
search user.net
nameserver 127.0.0.1
[root@server ~]#
```

Рис. 6: Добавление файла описания зоны в named.conf

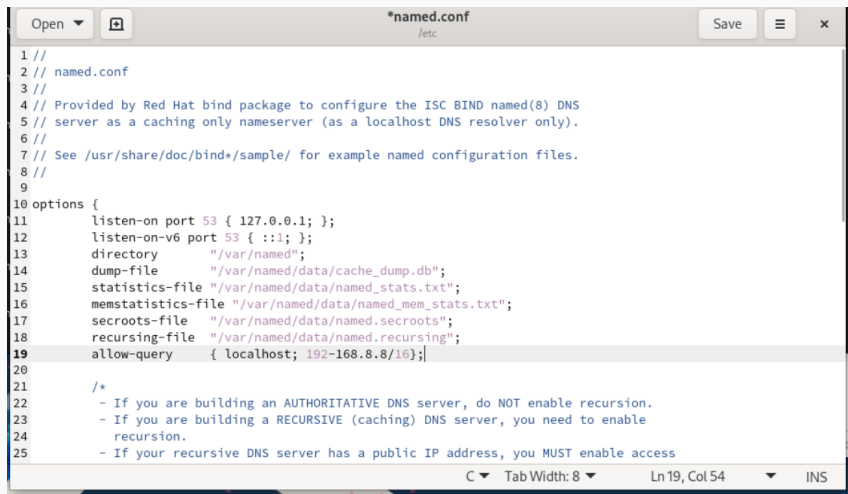
Создание прямой и обратной зон



```
1 //
2 // named.conf
3 //
4 // Provided by Red Hat bind package to configure the ISC BIND named(8) DNS
5 // server as a caching only nameserver (as a localhost DNS resolver only).
6 //
7 // See /usr/share/doc/bind*/sample/ for example named configuration files.
8 //
9
10 options {
11     listen-on port 53 { 127.0.0.1; };
12     listen-on-v6 port 53 { ::1; };
13     directory      "/var/named";
14     dump-file       "/var/named/data/cache_dump.db";
15     statistics-file  "/var/named/data/named_stats.txt";
16     memstatistics-file "/var/named/data/named_mem_stats.txt";
17     secroots-file   "/var/named/data/named.secroots";
18     recursing-file  "/var/named/data/named.recurse";
19     allow-query     { localhost; };
20
21     /*
22      - If you are building an AUTHORITATIVE DNS server, do NOT enable recursion.
23      - If you are building a RECURSIVE (caching) DNS server, you need to enable
24        recursion.
25      - If your recursive DNS server has a public IP address, you MUST enable access
```

Рис. 7: Описание прямой и обратной зон в файле akabrikosov.net

Создание файла прямой зоны



```
1 //
2 // named.conf
3 //
4 // Provided by Red Hat bind package to configure the ISC BIND named(8) DNS
5 // server as a caching only nameserver (as a localhost DNS resolver only).
6 //
7 // See /usr/share/doc/bind*/sample/ for example named configuration files.
8 //
9
10 options {
11     listen-on port 53 { 127.0.0.1; };
12     listen-on-v6 port 53 { ::1; };
13     directory      "/var/named";
14     dump-file       "/var/named/data/cache_dump.db";
15     statistics-file "/var/named/data/named_stats.txt";
16     memstatistics-file "/var/named/data/named_mem_stats.txt";
17     secroots-file   "/var/named/data/named.secroots";
18     recursing-file   "/var/named/data/named.recursing";
19     allow-query      { localhost; 192-168.8.8/16; };
20
21     /*
22      - If you are building an AUTHORITATIVE DNS server, do NOT enable recursion.
23      - If you are building a RECURSIVE (caching) DNS server, you need to enable
24        recursion.
25      - If your recursive DNS server has a public IP address, you MUST enable access
```

Рис. 8: Файл прямой зоны akabrikosov.net

Создание файла обратной зоны

```
[root@server ~]# cat /etc/resolv.conf
# Generated by NetworkManager
search user.net
nameserver 127.0.0.1
[root@server ~]# firewall-cmd --add-service=dns
success
[root@server ~]# firewall-cmd --add-service=dns --permanent
success
[root@server ~]# sof | grep UDP
bash: sof: command not found...
[root@server ~]# lsof | grep UDP
lsof: WARNING: can't stat() fuse.gvfsd-fuse file system /run/user/1000/gvfs
Output information may be incomplete.
lsof: WARNING: can't stat() fuse.portal file system /run/user/1000/doc
Output information may be incomplete.
```

Рис. 9: Файл обратной зоны 192.168.1

Настройка прав и контекстов SELinux

```
root@server:~  
named 3187 3240 isc-net-0 named 67u IPv6 34440 0t0 UDP lo  
calhost:domain  
named 3187 3240 isc-net-0 named 68u IPv6 34441 0t0 UDP lo  
calhost:domain  
named 3187 3240 isc-net-0 named 69u IPv6 34442 0t0 UDP lo  
calhost:domain  
named 3187 3241 isc-net-0 named 54u IPv4 33481 0t0 UDP lo  
calhost:domain  
named 3187 3241 isc-net-0 named 55u IPv4 33482 0t0 UDP lo  
calhost:domain  
named 3187 3241 isc-net-0 named 56u IPv4 33483 0t0 UDP lo  
calhost:domain  
named 3187 3241 isc-net-0 named 57u IPv4 33484 0t0 UDP lo  
calhost:domain  
named 3187 3241 isc-net-0 named 66u IPv6 34439 0t0 UDP lo  
calhost:domain  
named 3187 3241 isc-net-0 named 67u IPv6 34440 0t0 UDP lo  
calhost:domain  
named 3187 3241 isc-net-0 named 68u IPv6 34441 0t0 UDP lo  
calhost:domain  
named 3187 3241 isc-net-0 named 69u IPv6 34442 0t0 UDP lo  
calhost:domain  
NetworkMa 3312 root 27u IPv4 33731 0t0 UDP se  
rver.user.net:bootpc->_gateway:bootps  
NetworkMa 3312 3320 gmain root 27u IPv4 33731 0t0 UDP se  
rver.user.net:bootpc->_gateway:bootps  
NetworkMa 3312 3322 gdbus root 27u IPv4 33731 0t0 UDP se  
rver.user.net:bootpc->_gateway:bootps  
[root@server ~]#
```

Анализ работы DNS-сервера

Проверка через dig

```
Open ▾ [icon] *named.conf /etc Save [menu] [x]

1 //
2 // named.conf
3 //
4 // Provided by Red Hat bind package to configure the ISC BIND named(8) DNS
5 // server as a caching only nameserver (as a localhost DNS resolver only).
6 //
7 // See /usr/share/doc/bind*/sample/ for example named configuration files.
8 //
9
10 options {
11     listen-on port 53 { 127.0.0.1; };
12     listen-on-v6 port 53 { ::1; };
13     directory "/var/named";
14     dump-file "/var/named/data/cache_dump.db";
15     statistics-file "/var/named/data/named_stats.txt";
16     memstatistics-file "/var/named/data/named_mem_stats.txt";
17     secroots-file "/var/named/data/named.secroots";
18     recursing-file "/var/named/data/named.recursing";
19     allow-query { localhost; 192-168.8.8/16 };
20
21     /*
22      * - If you are building an AUTHORITATIVE DNS server, do NOT enable recursion.
23      * - If you are building a RECURSIVE (caching) DNS server, you need to enable
24      * recursion.
25      * - If your recursive DNS server has a public IP address, you MUST enable access
26      * control to limit queries to your legitimate users. Failing to do so will
27      * cause your server to become part of large scale DNS amplification
28      * attacks. Implementing BCP38 within your network would greatly
29      * reduce such attack surface
30      */
31     recursion yes;
32
33     dnssec-enable no;
34     dnssec-validation no
35
36     managed-keys-directory "/var/named/dynamic";
37     genin-directory "/usr/share/GenIP".
```

Проверка через host

```
53 zone "." IN {  
54     type hint;  
55     file "named.ca";  
56 };  
57  
58 include "/etc/named/user.net";  
59
```

C ▼ Tab Width: 8 ▼ Ln 34, Col 29 ▼ INS

Рис. 12: Проверка зоны командой `host -l` и `host -a`

Проверка прямой и обратной записи

```
etc/named
1 // named.rfc1912.zones:
2 //
3 // Provided by Red Hat caching-nameserver package
4 //
5 // ISC BIND named zone configuration for zones recommended by
6 // RFC 1912 section 4.1 : localhost TLDs and address zones
7 // and https://tools.ietf.org/html/rfc6303
8 // (c)2007 R W Franks
9 //
10 // See /usr/share/doc/bind*/sample/ for example named configuration files.
11 //
12 // Note: empty-zones-enable yes; option is default.
13 // If private ranges should be forwarded, add
14 // disable-empty-zone "."; into options
15 //
16
17 zone "user.net" IN {
18     type master;
19     file "master/fz/user.net";
20     allow-update { none; };
21 };
22
23 zone "1.168.192.in-addr.arpa" IN {
24     type master;
25     file "master/rz/192.168.1";
26     allow-update { none; };
27 };
```

Внесение изменений в окружение

Копирование конфигураций DNS в среду Vagrant

```
root@server:/var/named/master/fz
```

```
type master;
file "named.localhost";
allow-update { none; };
};

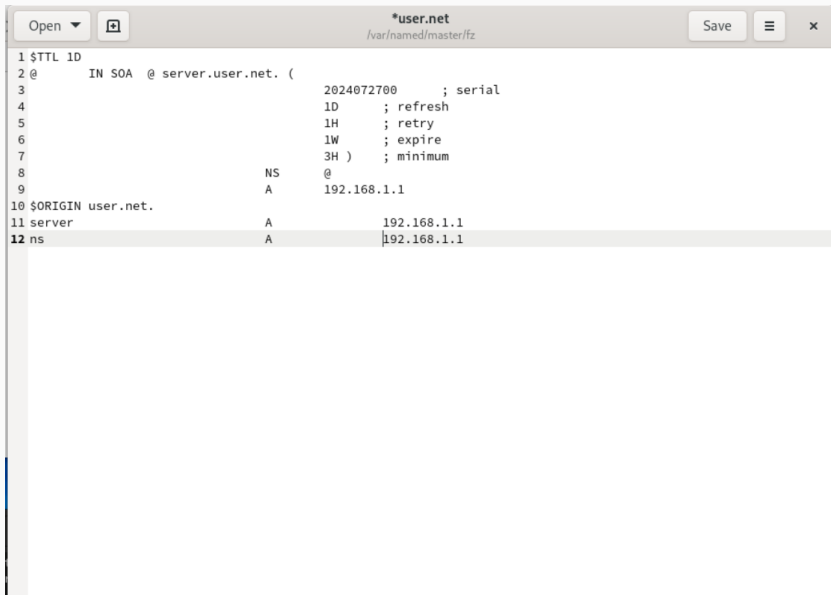
zone "1.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.ip6.arpa" IN {
    type master;
    file "named.loopback";
    allow-update { none; };
};

zone "1.0.0.127.in-addr.arpa" IN {
    type master;
    file "named.loopback";
    allow-update { none; };
};

zone "0.in-addr.arpa" IN {
    type master;
    file "named.empty";
    allow-update { none; };
};

[root@server named]# cd /var/named
[root@server named]# mkdir -p /var/named/master/fz
[root@server named]# mkdir -p /var/named/master/rz
[root@server named]# cp /var/named/named.localhost /var/named/master/fz/
[root@server named]# cd /var/named/master/fz/
[root@server fz]# mv named.localhost user.net
[root@server fz]#
```

Автоматизация конфигурации через dns.sh



The screenshot shows a text editor window titled '*user.net' with the file path '/var/named/masterffz'. The editor contains a DNS configuration file with the following content:

```
1 $TTL 1D
2 @      IN SOA  @ server.user.net. (
3                               2024072700      ; serial
4                               1D      ; refresh
5                               1H      ; retry
6                               1W      ; expire
7                               3H )   ; minimum
8                               NS      @
9                               A       192.168.1.1
10 $ORIGIN user.net.
11 server      A       192.168.1.1
12 ns          A       192.168.1.1
```

Выводы по проделанной работе

В результате лабораторной работы был установлен и настроен DNS-сервер с использованием **bind**.

Созданы прямые и обратные зоны домена **akabrikosov.net**, внесены необходимые записи.

Настроены права доступа и контексты SELinux, реализована автоматизация конфигурации через скрипт **dns.sh**.

Работоспособность DNS-сервера подтверждена тестами утилит **dig** и **host**.